Unit 2: Statistics

Definitions:

Statistics: is the branch of Mathematics concerned with the collection, interpretation and analysis of data.

Data: is the name given to the information collected on a particular subject, usually expressed in numerical form.

Data Handling involves:

- Collecting data.
- Sorting data.
- Representing data graphically (i.e. by bar graphs, pie-charts, broken line graphs, histograms and frequency polygons.)
- Analysing the data and interpreting the results.
- Making conclusions.

Classification of data:

We classify all data that we collect in two ways:

1. ***Quantitative/Numerical data****:* Data that can be discrete or continuous

   a) ***Discrete Data****:* This is information that can be collected by **counting**.
   For example:
   - The number of children in a family. This information can only exist as whole numbers and is counted.
   - Marks scored in a maths test. This data can include rational numbers and is also counted.

   b) ***Continuous Data****:* This information is collected by **measurement**. The data is given a suitable degree of accuracy but may not be given exactly.
   For example:
   - Heights of particular trees. The accuracy of the measurements will depend on the instruments being used.
   - Temperature measured at fixed intervals. Again the accuracy of the measurements will depend on the instruments being used.

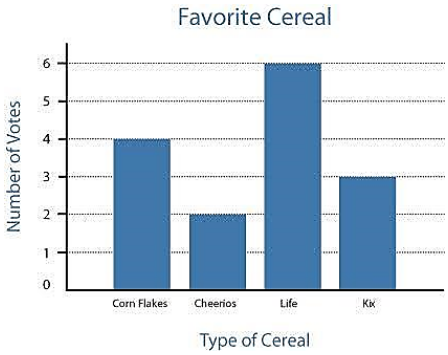
2. ***Qualitative/Categorical data****:* Data that is non-numeric. We can't find the mean or the median of qualitative data, but we can sometimes find the mode.
   For example: Gender; eye colours; favourite animal; cold drink types.

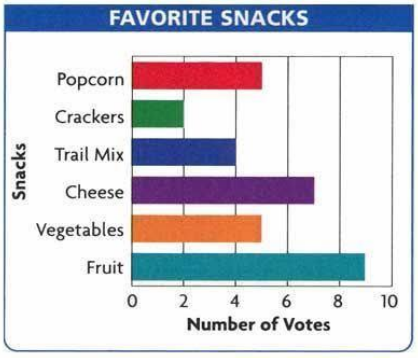
Representing data graphically:

Bar Graphs:

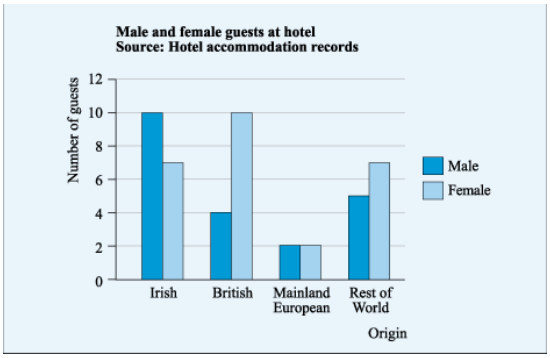
- A bar graph is a diagram consisting of a series of parallel bars (horizontal or vertical).
- The length/height of these bars show frequency, which makes the bars easy to compare.
- There is always a **space** between the bars.
- Bar graphs are used to display **categorical or discrete numerical data**. From this we get the frequency of the categories. E.g. the number of different coloured cars passing a point in a given time period; the number of people of a particular age (in full years).

- There are several types of bar graphs:
  - ***Normal Bar Graph*:** The graph is as described above and it is the most common of the bar graphs.

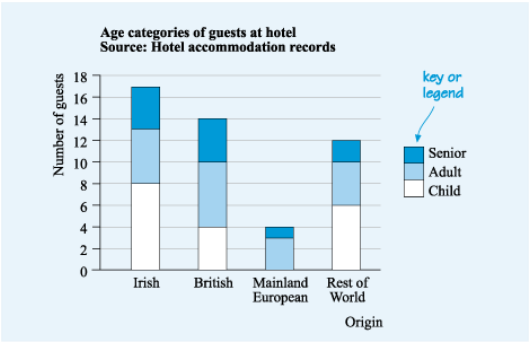




  - ***Compound bar graph*:** These are bar graphs that often have two bars next to one another, or bars stacked on top of one another. They use a difference in colour to distinguish the one from the other. They are generally used when different sets of information are given on connected topics.



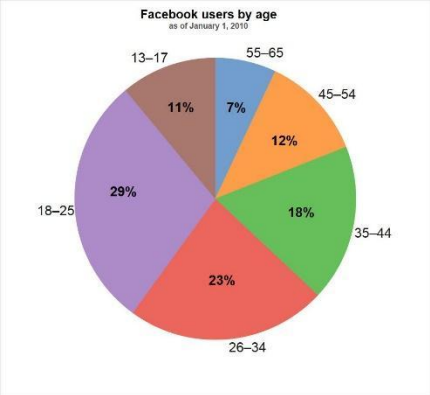
Dual Bar Graph

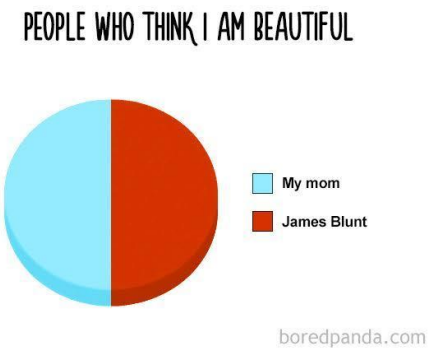


Sectional/Stacked Bar Graph

Pie Charts:

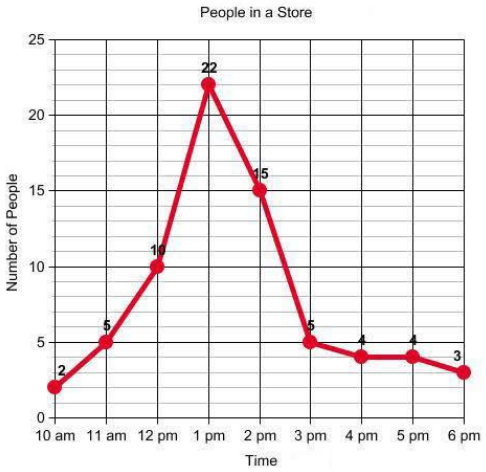
- Pie charts are useful when you want to illustrate how a population is divided up into different parts and what proportion of the whole each part represents.
- Pie charts are circular in shape and divided into sectors.
- The percentages per sector can be represented in degrees. To calculate the size of each slice of the pie chart we work out the angle of each slice by calculating it as a percentage of 360°.

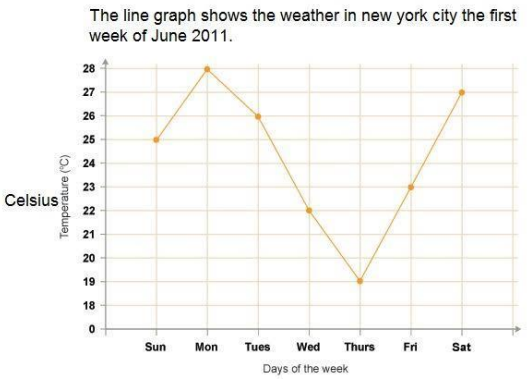




Broken Line Graphs:

A broken line graph is used to represent information collected at different points. It joins unique points and not every point on the line is an actual reading that has been plotted. They are used to show patterns or trends.





Summarising data:

Sometimes the quantity of data that one deals with is too large, making it necessary to **group** the data into what is known as **class intervals.** One way this can be done by means of a **stem-and leaf-diagram**. From there it can be plotted onto a **histogram** and/or **frequency polygon**.

Stem and Leaf Diagrams:

- This diagram is another way of ordering raw data.
- The stems represent the first digit of each number and the leaves the second. E.g. the stem can be the tens and the units the leaf, or the stem can be the units and the leaf the decimal place.

Race Running Times in Seconds

Stem	Leaves
12	2 6
13	0 2 5
14	1 2 4 6
15	2 3 7 8
16	1 2 4 6 8
17	5 7 8
18	1 3

Key: 14 \| 2 = 14.2 seconds

Phone Battery Comparison

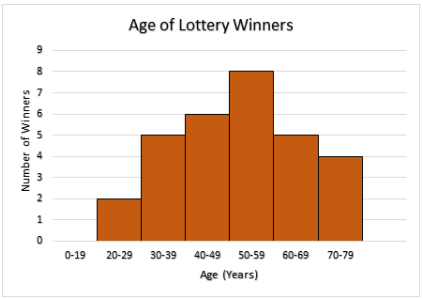
"Brand A" LEAF	STEM	"Brand B" LEAF
8 8 7 5	0	7
9 7 4 1 0	1	0 5 5 5 7 9
2 2 2 1	2	0 2 2 6 7
8 6 4 2 0	3	0 2 4 6 8
	4	
	5	6
1	6	
	7	5

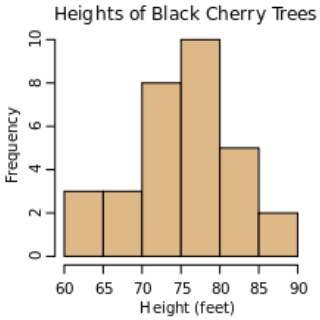
Key : 6 \| 1 = 61 hours

Histograms

- Histograms are generally used for grouped data that are divided into class intervals. It is similar to a bar graph but there are no spaces between the bars.
- Histograms are used to display **continuous numerical data**.
- Each bar represents a class interval and the frequency is read off from the vertical axis (height of each bar).

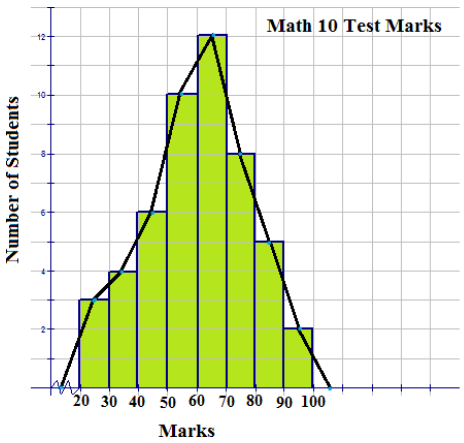
  Note: We must always leave an empty class interval before we start drawing the first bar.

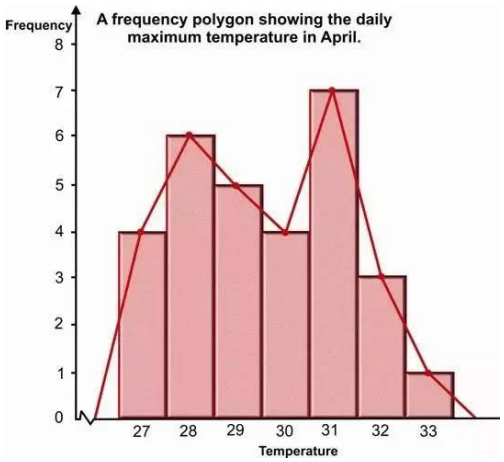




Frequency Polygons:

- A frequency polygon is a line graph in which the class mark of a particular interval is plotted against the frequency of that interval.
- It can often be constructed by joining the midpoints of the tops of the bars in a histogram.
- These always are “anchored” to (i.e. touch) the *x*-axis before and after the distribution of bars.





Biased data and the misuse of data in statistics:

It sometimes happens that statistical data is biased or misleading and it is therefore important that we analyse data to see where there are distortions or biases. Graphs can be misleading in the following ways:

- Using two or three dimensions in pictograms and bar charts (influences apparent/relative size).
- Using perspective or shape to exaggerate representation.
- Reversing the direction of the axis makes decreases look like increases.
- Altering the scale on an axis making the graph look much steeper or less steep than what it actually is.
- Leaving out part of an axis to exaggerate differences.

Homework: Ex. 1 pg. 350 – 351 (Mind Action Series new edition, 2022)

Measures of central tendency (ungrouped data):

Sometimes one wishes to compare the information from one set of data with that of another set of data. Looking at long lists or tables can sometimes be very confusing. If you are however, comparing one number with another, the information becomes easier to compare. The following all are types of averages for any given set of data.

The Mean:

We calculate the mean by **summing all the values in the data set and then dividing by the number of values** in the data set. This measure of central tendency is the most commonly used of all three averages. E.g. class average for a test.
The mean is easily affected by outliers (extremely high or extremely low values), and therefore is not always a reliable measure of central tendency.

The Median:

The median is the data point with 50% of the data values smaller than it (and therefore 50% of the data is larger in value too). The median is **middle value** (for an odd number of scores) or the **average of the two middle scores** (for an even number of scores). The scores should first be arranged in increasing order before the median can be determined.
The median is not affected by outliers and is therefore a good measure of central tendency.

The Mode:

The mode is the **most commonly occurring** observation in the data set i.e. the data point(s) with the highest frequency. There can be none, one, two (bimodal) or many (multimodal) modal values.
The mode is unaffected by outliers, but its position may not fall in the centre of the data (the mode is meant to be a measure of the centre), especially with data sets that contain more than one modal value.

Example 1:
The following information collected about the number of whatsapp messages sent by grade 10 learners in one day showed the following results:

25	36	55	19	58	14	44	36	28	69	58	15	28
31	22	58	26	38	22	17	34	50	29	12	25	32
36	58											

a) Calculate the mean number of WhatsApp messages sent per grade 10 learner in one day. Show your calculations.

b) Calculate the median of this data set?

c) What is the mode in this data set?

Example 2:
A teacher recorded the marks (out of 10) of 40 grade 10 learners. The data was recorded in a frequency table below:

Marks (x_i)	**Tally**	**Frequency (f)**	$x_i \times f$
1	||	2	
2	|||	3	
3	||||	4	
4	𝍸 𝍸	10	
5	𝍸 |	6	
6	||||	4	
7	𝍸	5	
8	|||	3	
9	|	1	
10	||	2	
TOTALS		40	

a) Determine the mean of the data set from the table.

b) Determine the mean of the data set from the table.

c) Determine the mode of the data set from the table.

Measures of central tendency (grouped data):

The Mean:

In order to find the *estimated mean* of grouped data, we have to find one value that can be used to represent each class. The fairest single value to use is the midpoint of each class.

The Median:

The Median Class would be the class that falls in the middle of the grouped data.

The Modal class:

The modal class is the class with the highest frequency. The estimated mode is considered to be the midpoint of the modal class interval.

Example 3:
David is investigating leaf sizes of different plants. The frequency table below shows the widths of the leaves on a Geranium plant.

Width of leaf (in mm)	11 – 20	21 – 30	31 – 40	41 – 50	51 – 60
Number of leaves	2	6	8	5	4

Calculate the **modal class** for the data above:

Calculate the **estimated mean** of the data below. Use the table below to assist you.

Width of leaf (in mm)	11 – 20,999… [11 ; 21)	21 – 30,999… [21 ; 31)	31 – 40,999… [31 ; 41)	41 – 50,999… [41 ; 51)	51 – 61 [51 ; 61]
Midpoint of class (m)					
Frequency (*f*)	2	6	8	5	4
$m \times f$					

Example 4: The following shows the heights of 48 grade 10 learners:

158	170	175	156	171	134	143	157	136	177	145	139
179	176	164	135	172	158	155	132	168	162	139	141
168	153	157	137	156	158	171	179	135	139	164	162
168	142	148	166	139	143	134	167	162	154	158	141

a) Decide on class intervals to group the data into.
b) Draw up a tally table of the height using your class intervals.

Class	**Tally**	**Frequency (*f*)**	**Midpoint of class interval**	**Frequency × midpoint**

d) Calculate the actual mean of the data.

e) Determine an estimate of the mean using the frequency table.

f) What is the mode in this data set?

g) What is the modal class?

h) Using the frequency table determine the median of this data set?

Homework: Ex. 2 pg. 356 – 357 (Mind Action Series new edition, 2022)
Ex. 6 pg. 375 – 376 (Mind Action Series new edition, 2022)

Measures of dispersion/spread (ungrouped data):

As well as knowing the central tendencies of a set of data, it is also important to have a measure of how data is spread/dispersed around the centre. This enables us to establish whether the data is grouped closely or scattered more widely (consistent or inconsistent). The measures of dispersion we will study (in Grade 10) are:

- Range
- Quartiles
- Interquartile range
- Semi-interquartile range
- Deciles
- Percentiles

The Range

The range is the difference between the largest and the smallest value in the data set. The bigger the range, the more spread out the data is.

$$\textbf{Range = Max. value – Min. value}$$

A great deal of information is ignored when computing the range, since only the largest and smallest data values are considered. The range of a data set is greatly influenced by the presence of just one unusually large or small value (outlier). As a result, other measures are required in order to give a better picture of the data spread. The range is an informative tool used as a supplement to other measures.

Quartiles

The quartiles divide the data into four equal sets:

- The first (lower) quartile is the value of the middle of the first set, where 25% of the values are smaller than Q_1 and 75% are larger. This first quartile is represented by the symbol Q_1.

$$\text{Position } (Q_1) = \frac{1}{4}(n+1)$$

- The second quartile (Q_2) is the median.

$$\text{Position } (Q_2) = \frac{2}{4}(n+1) = \frac{1}{2}(n+1)$$

- The third (upper) quartile is the value of the middle of the second set, where 75% of the values are smaller than Q_3 and 25% are larger. This third quartile is represented by the symbol Q_3.

$$\text{Position } (Q_3) = \frac{3}{4}(n+1)$$

Interquartile range:

The difference between upper and lower quartiles ($Q_3 - Q_1$), which is called the interquartile range, also indicates the dispersion of a data set. The interquartile range spans 50% of a data set, and eliminates the influence of outliers because, in effect, the highest and lowest quarters are removed.

Interquartile range (IQR) = $Q_3 - Q_1$

Example 5:

A year ago, Angela began working at a computer store. Her supervisor asked her to keep a record of the number of sales she made each month. The following data set is a list of her sales for the last 12 months:

34, 47, 1, 15, 57, 24, 20, 11, 19, 50, 28, 37.

Use Angela's sales records to determine:

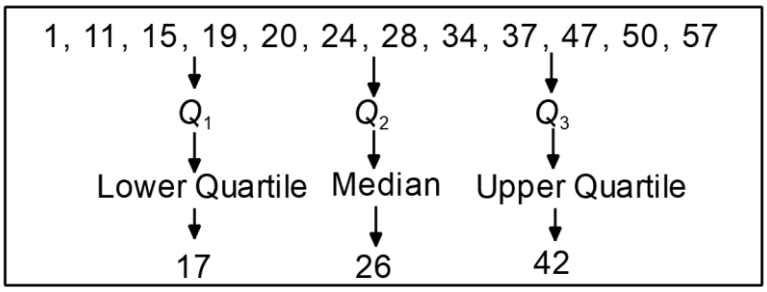
a) the median

b) the range

c) the upper and lower quartiles

d) the interquartile range

These results can be summarized as follows:



Note: This example has an even number of observations. The median, Q_2, lies between the centre of two observations (24 and 28), so the calculation of Q_1 includes the

observation 24 as it is below the value of Q_2. Similarly, 28 is also included in the calculation of Q_3 as it is above the value of Q_2.

Consider an odd number of observations such as l, 2, 3, 4, 5, 6, 7. Here the value of Q_2 is 4. As the location of the median is right on the fourth observation, this value is not included in calculating Q_1 and Q_3 , as we are interested only in the data above and below Q_2. In the above example, $Q_1 = 2$ and $Q_3 = 6$.

Example 6:

Alongside is a back-to-back stem and leaf plot for the marks obtained by two classes for the term.

Calculate the median, lower quartile, and upper quartile for each class.

Class 1		Class 2
0	10	
2	9	
8 6	8	0 2
6	7	8 8 8
6 4	6	0 2 6 6
4 2	5	6 8
8	4	
	3	
2	2	

Semi-interquartile range

The semi-inter-quartile range is sometimes quoted. It is half of the inter-quartile range.

$$\textbf{Semi-interquartile range} = \frac{Q_3 - Q_1}{2}$$

Example 7:

12 learners in a third class have term marks as follows:

20 39 40 43 43 46 53 58 63 70 75 91

Determine the Median, Range, Quartiles and Inter-quartile range.
Indicate in the space provided what each result infers.

Median		
Q_1 and Q_3		

Range		
Inter Quartile Range		
Semi-inter Quartile Range		

Deciles and Percentiles:

The **deciles** divide the data into tenths.

The **percentiles** divide the data into hundredths. The percentiles are simply called the 1st percentile, the 2nd percentile and so on.

The position of P_n, the nth percentile (the position of the data point with n% of the data smaller in value and (100 – n)% of the data larger in value than it) is determined by:

$$\text{Position } (P_n) = \frac{n}{100}(n+1)$$

Then, count the ordered data to locate the value in the calculated position.

Homework: Ex. 5 pg. 371 – 373 (Mind Action Series new edition, 2022)

Five Number Summaries and Box-and-Whisker Diagrams:

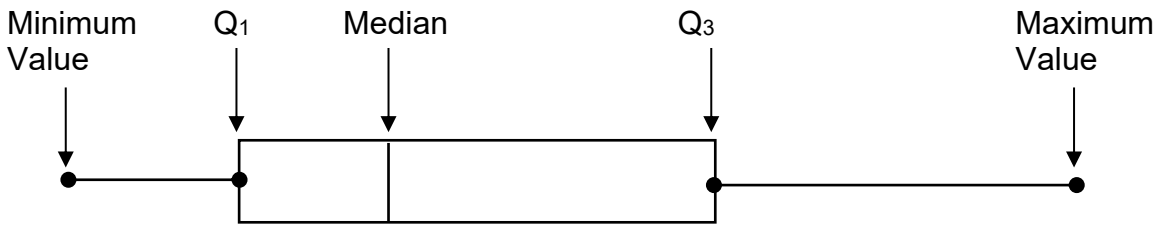
The Five Number Summary

These are the 5 important values needed to make a box-and-whisker diagram:

1. Minimum (smallest value)
2. First quartile (Q_1)
3. Median
4. Third Quartile (Q_3)
5. Maximum (largest value)

Box-and whisker diagrams

- A box-and-whisker diagram is an easy way to see how data is dispersed around the median.
- It is a graphical representation of the 5 Number Summary.



- The box shows where the values from the inter-quartile range lie (the middle 50% of the data).
- The whiskers extend from the two quartiles to the lowest and highest value.
- Exactly one quarter (25%) of the data lies in each section of the box and on each whisker.

You can draw a box-and-whisker by following 6 easy steps:

Example 8:

A farmer measured the height of 26 of his mealie plants in cm:

123	104	106	170	113	120	111	119	121	110	124	116	100
130	124	123	127	115	116	120	111	109	106	106	107	101

Draw a box-and-whisker diagram to display this information:

We have to organise the data first:
KEY:

Stem	Leaves	Ordered leaves
10		
11		
12		
13		
14		
15		
16		
17		

Step 1: Determine the Five Number Summary

Step 2:
- Draw an axis which covers the entire range of the data.
- Remember to leave space above it for your box-and-whisker diagram.
- Ensure that your scale is correct, just like a number line (like a ruler).
- Ensure that your number line covers all the values from bottom to top.

Step 3: Draw vertical lines above the values of Q1, Q2(median) and Q3.

Step 4: Join the lines on the top & the bottom to form a box.

Step 5: Extend a line from the LHS of the box to the minimum value. Extend a line from the RHS of the box to the maximum value.

Step 6: Give your diagram a heading.

Skewed & Symmetrical Distributions:

- Symmetrical data describes data that is balanced in relation to the mean or median.
- Skewed data describes data that is clustered more on one side of the mean or median than the other.

When looking at skewness, we can compare the mean and the median of a data set:

- If **mean ≈ median** i.e. very close, the data is **symmetrical**



- If **mean > median** the data is **skewed to the RIGHT** (also called **POSITIVELY SKEWED**)



- If **mean < median** the data is **skewed to the LEFT** (also called **NEGATIVELY SKEWED**)



Homework: Ex. 3 pg. 363 – 364 (Mind Action Series new edition, 2022)

Outliers:

Outliers are data points that are significantly far away from all of the other data points. Outliers could be either significantly small or significantly large values.
These 'singularities' are unexpected values when compared to the rest of the data, e.g. an orange that weighs 1 kg or an orange that weighs 70 g; these are unexpected masses for oranges, and each really would "stick out" in a random sample of oranges.
The min. and max. values are NOT 'automatically' outliers – this is important.
Not all data sets have outliers – this also is important.

Homework: Ex. 4 pg. 367 – 369 (Mind Action Series new edition, 2022)
Homework: Consolidation and extension exercise pg. 377 – 381